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FCS Live Session

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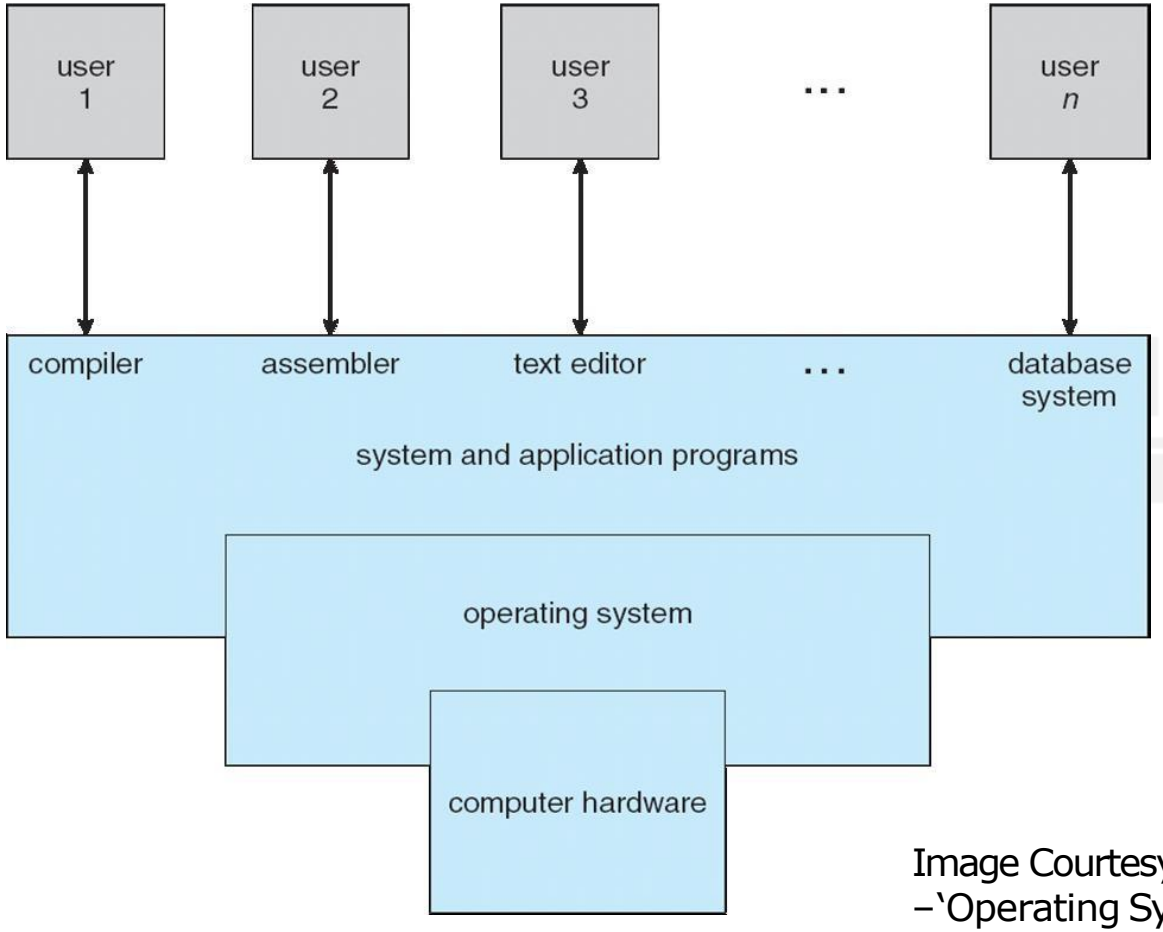
Computer System Structure

Computer system can be divided into three components.

- Hardware – provides basic computing resources.
- Software – Set of programs for data processing.
  System software(OS) and application software.
- Users – People, machines or other computers.

Computer System Structure



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Classification of OS

Classification based on the concept of process.

Process

- A program in execution is called a process.
- When a process is on, it will continuously need various types of resources.
- It will need CPU, I/O devices, main memory and various types of software resources.

Classification of OS

Single-tasking vs Multi-tasking

- A single-tasking system can only run one process at a time, while a multi-tasking operating system allows more than one process to be running in concurrency
- This is achieved by time-sharing, where the available processor time is divided between multiple processes.
- Other resources too would need to be shared in a multi-tasking environment.

Classification of OS

Single-user vs Multi-user

- Single-user operating systems have no facilities to distinguish users, but may allow multiple processes to run in tandem
- A multi-user operating system extends the basic concept of multi-tasking with facilities that identify processes and resources
- The system permits multiple users to interact with the system at the same time (often via 'dumb' terminals).

Classification of OS

Distributed Operating systems

- A distributed system is a collection of physically separate, possibly heterogeneous, computer systems that are networked to provide users with access to the various resources that the system maintains.
- Appear to be a single computer, as all computations are distributed - divided amongst the constituent computers.

Classification of OS

Embedded Operating Systems

- Embedded systems are computers which control some larger machinery.
- Embedded operating systems are designed to be used in embedded computer systems.
- They are designed to operate on small machines with less autonomy.
- They are very compact and extremely efficient by design, and are able to operate with a limited amount of resources.

Classification of OS

Real-time Operating System

- Real time computers operate with stringent real time requirements.
- A real-time operating system is an operating system that guarantees to process events or data by a specific moment in time.
- Such an OS may be single- or multi-tasking.
- specialized scheduling algorithms for multi-tasking.

OS Operations

- **Interrupt driven** (hardware and software)
  - Hardware interrupt.
  - Software interrupt (**exception** or **trap):**
    - Software error (e.g., division by zero)
    - Request for operating system service.
    - Other process problems.

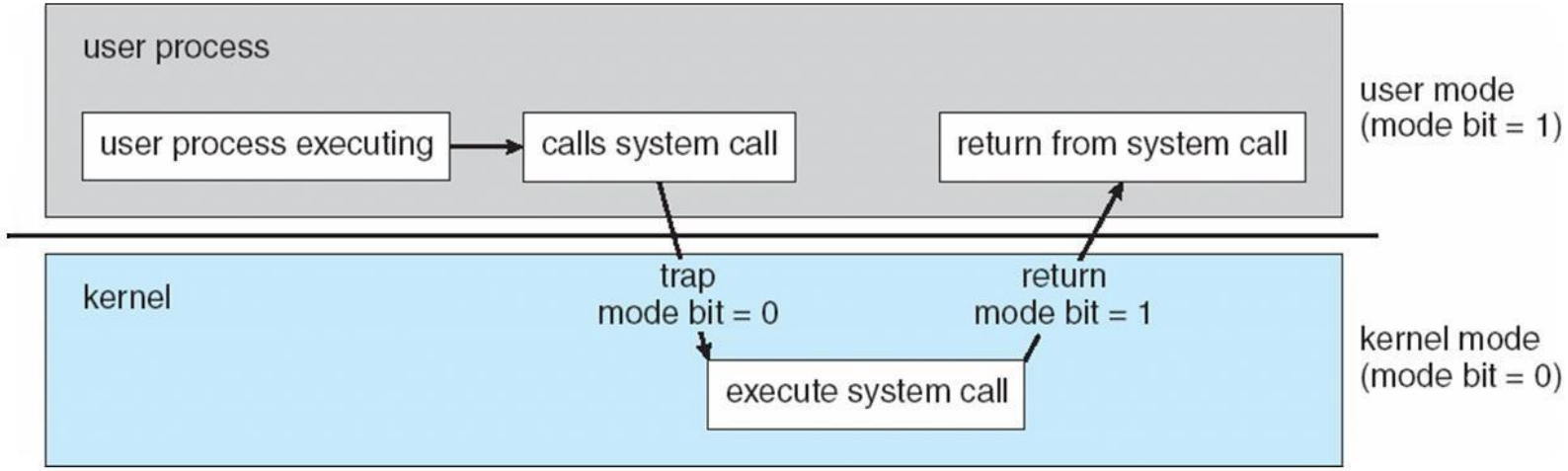
    Eg. infinite loop, processes modifying each other or the operating system.

OS Operations

- **Dual-mode** operation allows OS to protect itself and other system components.
  - **User mode** and **kernel mode**
  - **Mode bit** provided by hardware.
    - Provides ability to distinguish when system is running user code or kernel code.
    - Some instructions designated as **privileged**, only executable in kernel mode.
    - System call changes mode to kernel, return from call resets it to user.

Transition from user mode to kernel

- Timer to prevent infinite loop / process hogging resources.
- When counter zero generate an interrupt.



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Functions of OS

- Process Management
- Memory management
- Peripheral devices management
- Protection and security
- Networking and communication

Memory

- Memory means any data storage mechanism.
- Need efficient management of how data is stored, accessed and manipulated.
- The storage and access are taken care of by the memory management module of the OS.

Memory Hierarchy

- Computer memory is usually organized hierarchically –in various well defined levels.

There are four major storage levels in the computer memory

- Internal –Processor registers and cache.
- Main –the system RAM and controller cards.
- On-line mass storage –Secondary storage.
- Off-line bulk storage –Tertiary and Off-line storage.

Memory

Memory Hierarchy

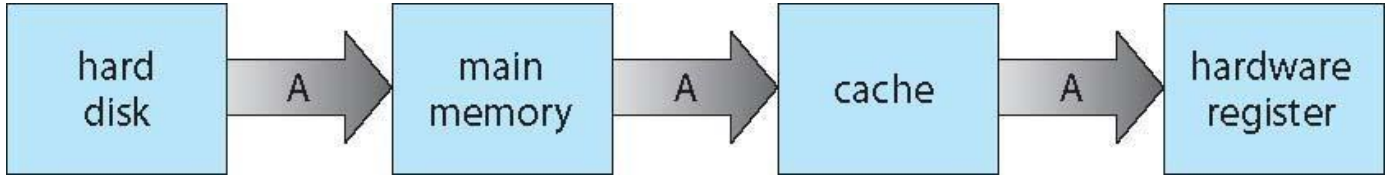
Level	1	2	3	4	5
Name	registers	cache	main memory	solid state disk	magnetic disk
Typical size	< 1 KB	< 16MB	< 64GB	< 1 TB	< 10 TB
Implementation technology	custom memory with multiple ports CMOS	on-chip or off-chip CMOS SRAM	CMOS SRAM	flash memory	magnetic disk
Access time (ns)	0.25 - 0.5	0.5 - 25	80 - 250	25,000 - 50,000	5,000,000
Bandwidth (MB/sec)	20,000 - 100,000	5,000 - 10,000	1,000 - 5,000	500	20 - 150
Managed by	compiler	hardware	operating system	operating system	operating system
Backed by	cache	main memory	disk	disk	disk or tape

Memory Hierarchy

- Register (very small capacity, temporary, extremely fast).
- Cache (small capacity, temporary, very fast).
- Main memory (medium capacity, temporary, fast).
- Secondary Memory (hard disk) (high capacity, permanent, not fast).
- Tertiary memory (very high capacity, permanent, slow).

Data Migration

- OS focussing on data movement between secondary and main memory.
- Microprocessor hardware handles data movements between main memory and internal storage.



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Memory Management

- To execute a program all (or part) of the instructions must be in memory
- Memory management determines what is in memory and when.
- Optimizing CPU utilization and computer response to users.
- Memory management activities
  - Keeping track of which parts of memory are currently being used and by whom.
  - Deciding which processes (or parts thereof) and data to move into and out of memory.
  - Allocating and deallocating memory space as needed.

Software Components

- A set of programs.
- Three types of software.
  - System Software
  - Application Software
  - Utility software

Software Components

System Software

- Controls all the components of a computer.
- Controls other software and hardware components.

Application software

- It is designed for a specific purpose of the user.
- Dedicated to a specific application.
- Run by users.
- Utility software is designed to solve any issues or a complex situation in a computer.

What is an Operating System(OS)

- An operating system (OS) is system software that manages all resources of a computer system.
- Controls Computer hardware including the core computer and peripheral devices, software resources.
- Provides common services for computer programs.

What is an Operating System(OS)

- Acts as an intermediary between user and the computer hardware.
- OS is a **resource allocator**
  - Manages all resources.
  - Decides between conflicting requests for efficient and fair resource use.
- OS is a **control program**
  - Controls execution of programs to prevent errors and improper use of the computer.

Basic Concepts

Some of the key concepts:

- Kernel
- System Calls
- Interrupts
- Shell

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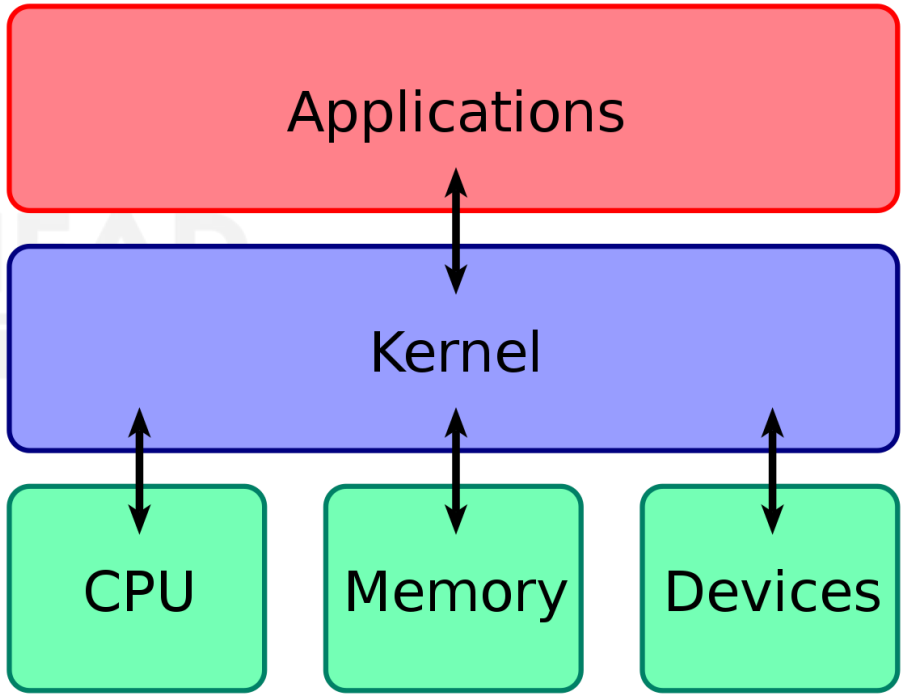
Kernel

- The kernel is the core component of an operating system.
- It is the part of the operating system that loads first, and it remains in main memory.
- It is responsible for providing secure access to the system's resources.
- It act as an interface between the user applications and the hardware.
- When a process makes requests of the kernel, the request is called **system call.**



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System Calls

- A system call is the means by which a computer program requests a service from the kernel of the operating system on which it is executed.
- This may include hardware-related services (for example, accessing a hard disk drive), creation and execution of new processes, and communication with integral kernel services such as process scheduling.
- System calls provide an essential interface between a process and the operating system.

Interrupts

- An interrupt is a response by the processor to an event that needs attention from the software
- An interrupt condition alerts the processor and serves as a request for the processor to interrupt the currently executing code when permitted, so that the event can be processed in a timely manner
- If the request is accepted, the processor responds by suspending its current activities, saving its state, and executing a function called an interrupt handler (or an interrupt service routine, ISR) to deal with the event

Shell

- A shell is a user interface for access to an operating system's services.
- In general, operating system shells use either a command-line interface (CLI) or graphical user interface (GUI), depending on a computer's role and particular operation.

Shell

Types of Shells

- A GUI (graphical user interface) - most famously seen in Windows family operating systems
- Command line interface – usually associated with Unix family
- Command-line shells require the user to be familiar with commands and their calling syntax, and to understand concepts about the shell-specific scripting language

Takeaways

- Computer System Structure.
- Software Components.
- What is an Operating System?

References

1 Silberschatz, Galvin and Gagne –'Operating System concepts' – 9th Edition.

2 J. Glenn Brookshear,"Computer Science: An Overview", Addision-Wesley, Twelth Edition, 2014

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